Hard Rods and Cavity Function

Complex disordered systems: Colloids

Consider a 1-dimensional (1D) system of **Hard Rods** of length σ. The number density is ρ, and the equation of state is exactly known as (Tonks equation): $Z_{\text{comp}} = \frac{1}{1-\rho\sigma}$.

The potential is:

$$u(x) = \begin{cases} \infty, & x < \sigma \\ 0, & x \geq \sigma \end{cases}$$

The **pair distribution function at contact**, $g(\sigma^+)$, is known to be directly related to the equation of state by the 1D virial theorem (see lecture notes):

$$Z = 1 + \rho\sigma g(\sigma^+)$$

1. Calculate $g(\sigma^+)$ for the 1D hard rod system using the Tonks equation of state and the 1D virial theorem.
2. Determine the value of the Boltzmann factor, $e^{-\beta u(\sigma)}$, at contact.
3. Determine the value of the cavity function, $y(r) = e^{\beta u(r)} g(r)$, at contact.
4. Explain the physical meaning of $y(\sigma)$. How does it differ from $g(\sigma^+)$ in its interpretation?
5. Derive $g(r)$ for $\sigma \leq r < 2\sigma$ by maximising the (Shannon) entropy

   $$S[p] = -\int_0^\infty p(l) \ln p(l) dl$$

   for the probability to find a gap of length l between neighbouring rods. Assume that the gaps are independent (for hard rods this is in fact exact) and check that the $g(r)$ you obtain matches the contact value found above.

Solution

1. *Calculate $g(\sigma^+)$*

We use the Tonks equation and the virial theorem:

$$Z_{\text{comp}} = \frac{1}{1-\rho\sigma}$$

$$Z_{\text{comp}} = 1 + \rho\sigma g(\sigma^+)$$

Setting the two expressions for Z equal:

$$1 + \rho\sigma g(\sigma^+) = \frac{1}{1-\rho\sigma}$$

Isolating $g(\sigma^+)$:

$$\rho\sigma g(\sigma^+) = \frac{1}{1-\rho\sigma} - 1$$

Combine the right side using a common denominator:

$$\rho\sigma g(\sigma^+) = \frac{1-(1-\rho\sigma)}{1-\rho\sigma} = \frac{\rho\sigma}{1-\rho\sigma}$$

Dividing by $\rho\sigma$:

$$g(\sigma^+) = \frac{1}{1-\rho\sigma}$$

This is the exact, analytic expression for the contact value for 1D hard rods.

2. *Determine* $e^{-\beta u(\sigma)}$

The Boltzmann factor $e^{-\beta u(x)}$ at contact $x = \sigma$ is determined by the potential $u(x)$:

$$e^{-\beta u(\sigma)} = e^{-\beta(0)} = 1$$

Since the potential is zero for $x \geq \sigma$, the particles are not interacting via the direct core repulsion precisely at σ^+.

3. *Determine* $y(\sigma)$

The cavity function $y(x)$ is related to $g(x)$ and the Boltzmann factor:

$$g(x) = e^{-\beta u(x)} y(x)$$

At contact, $x = \sigma$:

$$g(\sigma^+) = e^{-\beta u(\sigma)} y(\sigma)$$

Substituting the results from (1) and (2):

$$\frac{1}{1-\rho\sigma} = (1) \cdot y(\sigma)$$

Therefore, the cavity function at contact is:

$$y(\sigma) = \frac{1}{1-\rho\sigma}$$

In this 1D case, $g(\sigma^+)$ and $y(\sigma)$ are equal.

4. Explain the meaning of $y(\sigma)$

Let's think of the direct correlation function

It is useful to think about the Percus–Yevick approximation for the direct correlation function

$$c(r) = (1 - e^{+\beta u(r)})(h(r) + 1)$$

(see lecture notes).

This can be rewritten as

$$c(r) = (1 - e^{+\beta u(r)})g(r) = (1 - e^{+\beta u(r)})y(r)e^{-\beta u(r)}$$

and then

$$c(r) = (e^{-\beta u(r)} - 1)y(r) = f(r)y(r)$$

where we used the Mayer f-function. While $f(r)$ jumps from -1 to 0 at contact, $y(r)$ is continuous, which makes the Ornstein–Zernike problem tractable. In particular, $y(r)$ removes the discontinuities caused by the potential at short distances and focuses on the correlations produced outside of the core. $g(\sigma^+)$ measures the **total** probability density, relative to an ideal gas, of finding a particle one rod length (σ) away from another, including both the direct repulsion *and* the indirect correlations (the rest of the fluid). It is an **experimental, measurable** quantity directly linked to pressure via the virial theorem.

The cavity function $y(r)$ is the probability of finding a particle at distance σ **if the direct, instantaneous repulsive interaction** $u(x)$ **were suddenly removed** (i.e., if $u(\sigma)$ was momentarily zero). It represents the correlation due **only to the structure induced by the surrounding fluid**.
We can reread the PY approximation in this light

$$c(r) = g_{\text{total}} - g_{\text{indirect}}$$

We assume that there is a potential of mean force $w(r)$ generating the correlations, such that

$$g(r) = g_{\text{total}}(r) = e^{-\beta w(r)}$$

and then we decide to **approximate** the indirect part by **removing** the interparticle potential from the potential of mean force

$$g_{\text{indirect}} = e^{-\beta(w(r)-u(r))}$$

Then the PY approximation with the $c(r) = f(r)y(r)$ comes about.

5. *Derivation*

Hard rods **cannot overlap**, so:

$$g(r) = 0 \quad (r < \sigma).$$

Let the free gap between rods be

$$l = r - \sigma \geq 0.$$

We use the maximum entropy principle to find $p(l)$.
We want to maximise

$$S[p] = -\int_0^\infty p(l) \ln p(l) dl$$

subject to

$$\int_0^\infty p(l)dl = 1, \quad \int_0^\infty lp(l)dl = \lambda$$

where the mean free gap is

$$\lambda = \frac{1 - \rho\sigma}{\rho}$$

Via Lagrange multipliers

$$\delta\left[-\int p \ln p dl + \alpha\left(\int p dl - 1\right) + \gamma\left(\int lp dl - \lambda\right)\right] = 0$$

and then

$$p(l) = \frac{1}{\lambda} e^{-l/\lambda}$$

and

$$\gamma = -\frac{1}{\lambda}, \quad \alpha = 1 - \ln\lambda$$

We then convert from gap probability to pair correlation.
For $\sigma \leq r < 2\sigma$ only the nearest neighbour can be at distance r, so the pair correlation is the gap distribution divided by the density:

$$g(r) = \frac{p(r-\sigma)}{\rho}, \qquad \sigma \leq r < 2\sigma.$$

Substitute $p(l)$ and λ:

$$g(r) = \frac{1}{(1-\rho\sigma)} \exp\left[-\frac{\rho(r-\sigma)}{1-\rho\sigma}\right], \qquad \sigma \le r < 2\sigma.$$

At larger r the second, third, … neighbours also contribute, each with the distribution of a sum of independent gaps, and $g(r)$ oscillates before decaying to 1.
Checks:

$$g(\sigma^+) = \frac{1}{1-\rho\sigma}.$$

Matches the 1D hard-rod equation of state:

$$P = \frac{\rho k_B T}{1-\rho\sigma}.$$